

EXP.NO:10

DATE:02.05.2025

USER INTERFACE DESIGN

Roll no:230701199

AIM:

Create a weather data visualization dashboard using JavaScript and Chart.js

PROCEDURE:

Step 1: Setup Html & Styles

HTML & CSS:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Inventory Management Visualization</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      text-align: center;
      margin: 50px;
      background-color: #f0f4f8;
    }
    canvas {
      margin: 30px auto;
      max-width: 500px;
    }
    h1 {
      color: #333;
    }
  </style>
</head>
<body>
  <h1>Inventory Management System</h1>

  <canvas id="pieChart" width="400" height="400"></canvas>
  <canvas id="barChart" width="400" height="400"></canvas>

  <!-- Chart.js Library -->
  <script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
  <!-- JavaScript Logic -->
  <script src="script.js"></script>
</body>
</html>
```

Step 2: JavaScript File (weather.js)

```
1  // script.js
2
3  // Inventory categories and stock data
4  const inventoryData = {
5      labels: ['Electronics', 'Clothing', 'Home Appliances', 'Books', 'Toys'],
6      datasets: [{
7          label: 'Items in Stock',
8          data: [200, 150, 100, 80, 50],
9          backgroundColor: [
10             '#FF6384', // Electronics
11             '#36A2EB', // Clothing
12             '#FFCE56', // Home Appliances
13             '#4BC0C0', // Books
14             '#9966FF'  // Toys
15         ]
16     }]
17 };
18
19 // PIE CHART
20 const ctxPie = document.getElementById('pieChart').getContext('2d');
21 const pieChart = new Chart(ctxPie, {
22     type: 'pie',
23     data: inventoryData,
24     options: {
25         responsive: true,
26         plugins: {
27             title: {
28                 display: true,
29                 text: 'Inventory Distribution'
30             }
31         }
32     }
33 });
34
35 // BAR CHART
36 const ctxBar = document.getElementById('barChart').getContext('2d');
37 const barChart = new Chart(ctxBar, {
```

```
37  const barChart = new Chart(ctxBar, {
38      type: 'bar',
39      data: inventoryData,
40      options: {
41          responsive: true,
42          plugins: {
43              title: {
44                  display: true,
45                  text: 'Items in Stock by Category'
46              }
47          },
48          scales: {
49              y: {
50                  beginAtZero: true
51              }
52          }
53      }
54  });
55
```

60 Hz

Output:

